

Ethan K. Long

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Summary

Data analyst specializing in sports analytics and predictive modeling, with experience using Python, SQL, and Tableau to deliver decision-support insights for NCAA Division I hockey and FDA-scale biomedical datasets. Strong foundation in statistical learning, feature engineering, and model validation, with demonstrated ability to translate complex data into actionable stakeholder decisions.

Education

University of Connecticut

Expected May 2026

Bachelor of Arts, Statistics

- **Coursework:** Intro to Data Science | Statistical Programming | Deep Learning | Mathematical Statistics | Multivariable Calculus

Academic Research Experience

Data Analyst Intern | UConn Men's Ice Hockey Team (NCAA Division I)

Aug. 2025 - Present

- Co-designed a custom event-logging framework for in-game breakout sequences, generating a labeled dataset of 1,000+ observations across the last two seasons.
- Identified breakdown patterns in breakout execution under defensive pressure, directly informing matchup-specific strategies and reducing reliance on low-success breakout sequences.

Research Assistant | Statistics Department, University of Connecticut

Apr. 2025 – Aug. 2025

- Developed end-to-end binary classification modeling on FDA-provided genomic datasets (80,000+ records), predicting phosphorylation events from codon-level and gene-level features under severe class imbalance.
- Engineered 115+ features from raw biological sequences, including selective one-hot encodings and derived codon statistics, while eliminating leakage and multicollinearity through sequential feature selection and correlation analysis.
- Built and validated ensemble models (SVM, RF, XGBoost) using weighted voting, improving performance from 63% → 87% F1 and 82% → 95% ROC-AUC.
- Reduced end-to-end runtime by ~35% through vectorization of gene sequences and aggressive featurizing pruning without degrading predictive performance.
- Maintained and shared, version-controlled repository reviewed by academic and FDA collaborators; presented findings biweekly to FDA research principals and weekly to faculty advisors.

Vice President | UConn Men's Club Ice Hockey Team

Sept. 2024 – Present

- Oversaw operations and a \$55,000 annual budget for a 30+ member organization, streamlining logistics and expense tracking.

Academic Projects

NYC Crash Data Analysis

- Cleaned and processed 500,000+ crash records using Python (Pandas, NumPy), focusing on human and environmental injury predictors.
- Conducted statistical correlation and hypothesis testing to isolate high-risk demographic, temporal, and spatial factors
- Built interactive Tableau dashboards to visualize crash risk patterns, enabling identification of high-risk demographic and geographic segments for targeted safety interventions.

Analysis of Narrative Bias in M&A Comparables Using 10-K Disclosures

- Constructed time-aligned corpora of acquiring-firm 10-k and proxy filings, enforcing strict event-window controls to prevent forward-looking contamination.
- Extracted and evaluated comparable company lists and associated valuation multiples, applying paired statistical tests to identify systematic difference between narrative selected and size/sector matched baselines.
- Demonstrated that narrative framing in disclosures can materially influence perceived valuation fairness, with implications for investors and deal evaluation.

ML-Based Evaluation of Clutch Performance in MLB Players

- Quantified “clutch” as performance degradation between regular and postseason play, modeling Total Bases vs. At-Bats to assess predictability loss under high-leverage constraints.
- Applied K-Means clustering to segment players into three postseason performance archetypes, validating cluster stability via elbow method and perturbation testing.
- Presented findings at the Connecticut Sports Analytics Symposium (2025), translating technical findings into actionable player evaluation insights.

Technical Skills

- Languages & Libraries: Python (pandas, numpy, scikit-learn), SQL (PostgreSQL, MySQL), R, SAS
- Quantitative Methods: Statistical testing, Regression, Clustering, Time series, Data cleaning, EDA
- Tools: Git, Docker, Jupyter, Tableau, Web scraping (BeautifulSoup, Selenium)